DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**BLOOD EXCHANGE TRANSFUSION**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-NICU-002 (v2)**		
Last Version Approval Date	**28/1/2022**	Number of Pages	**8**

	NAME	POSITION	SIGNATURE	DATE
Prepared by:	**MS. JOICY JOY**	NICU Quality Representative		27/2/2024
	MS. AISHA ZAIBI SAAD AL-SHUMRANI	Neonatal Intensive Care Unit, Head Nurse		27/2/2024
	DR. TAYSEER HASSAN	Neonatology Resident		27/02/2024
Reviewed by:	**DRA. FAWZIA MOHAMMED FARAHAT**	Head, Neonatology and Pediatric Department		14/03/24
	MR. MANSOUR AHMED AL-ZAHRANI	Head, Quality Management and Patient Safety		14-3-24
Approved by:	**MR. SULTAN ALI AL-GAMDI**	Nursing Director		19-3-24
	DR. WAEL KAMEL ARBA	HED Assistant for Medical Affairs		19.3.24

APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19/ 3 / 2024	7 / 4 / 2024	TRIENNIAL	18/ 3 / 2027

Baha_Cluster
Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

1.0 PURPOSE

1.1. To prevent accumulation of bilirubin in the blood above dangerous level.

1.2. To prevent kernicterus as well as to prevent accumulation of other by products of hemolysis from hemolytic disease.

1.3. To replace red blood cells which have poor oxygen releasing capacity and poor carbonic anhydrase activity as in premature infants.

1.4. To remove toxic metabolites and to correct anaemia.

2.0 DEFINITION/ABBREVIATION

2.1. **Exchange Transfusion** is a technique or procedure used most often to maintain serum bilirubin at levels below neurotoxicity. It refers to giving whole blood in exchange of an infant blood. The infant blood is repeatedly drawn out in small amount and replaced with equal amount of compatible donor blood. It is a potentially life-saving procedure that is done to prevent mortality and morbidity effects of serious diseases.

3.0 RESPONSIBILITY

3.1. **NICU Physician**:

3.1.1. Ordering basic investigation including CBC, RFT, LFT, HB Electrophoresis and ferritin.

3.1.2. Prepare all paper required for procedure including admission sheet and consent.

3.1.3. Ordering for blood exchange and determine product type, amount of red blood cell & estimate time for exchange.

3.1.4. Write clear plan for procedure.

3.1.5. Gives the parents detailed explanation about the procedure.

3.1.6. Observe patient pre, during and post procedure.

3.2. **NICU Nurse**:

3.2.1. Confirms the physician's order, signed consent, and patient identification accurately.

3.2.2. Coordinate with the blood bank for laboratory requests and cross matching.

3.2.3. Monitors patient's vital signs.

3.2.4. Checks the red blood cells accurately with physician workers before starting procedure to the patient.

3.2.5. Safe handling and storage of received blood.

3.2.6. Safe administration of blood.

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Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

3.2.7. To do blood exchange procedure under control and continuous observation of physician.

3.2.8. Monitors patient's vital signs during procedure.

3.2.9. Reports any untoward events during the procedure to the physician.

3.2.10. Cares for the patient during the course of the whole procedure.

4.0 POLICY

4.1. The infant's blood is repeatedly drawn out by the physician under strict aseptic technique in small amount and replaced with equal amount of compatible donor blood, assisted by a nurse with continuous monitoring of the newborn.

4.2. Consent for blood exchange must be explained by Physician and signed by parents for the newborn.

4.3. Continuous phototherapy should be given throughout the procedure and ensure adequate hydration throughout the process.

4.4. Standard precautions are to be followed when handling blood products.

4.5. Blood for exchange transfusion must be cross matched to the recipient's blood type, and two staff nurses should verify the patient's identity prior to blood extraction for type and cross match.

4.6. Type and cross match blood, will be Red Blood Cell, Group O negative, CMV negative, sickle cell negative, irradiated, leukocyte reduced, less than 10 days old, suspended in AB frozen plasma (or infant's type) and have a haematocrit between 45 - 50 should be used for exchange transfusion.

4.7. **Blood to be used for blood exchange must be counterchecked by blood bank technician and the nurse who will receive the blood**:

4.7.1. Name of patient and medical record number

4.7.2. Check blood for clots, hemolysis and leak on the blood bag.

4.7.3. **Confirm serial number on the blood bag level**:

4.7.3.1. Expiry date

4.7.3.2. Blood Group compatibility

4.7.3.3. Serology – negative results

4.8. Blood must be counterchecked by the attending Nurse and/ or doctor, once the blood is obtained from the blood bank.

4.9. Before starting blood exchange two nurses should identify the patient correctly using the appropriate means of identification such as using identification bracelet, with patient's name, medical record number, age, nationality, and asking the patient's name.

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4.10. Blood for exchange transfusion must be checked at patient's bedside by the nurse and/or a physician before starting the procedure

4.11. Baseline vital signs should be taken and recorded. Connect the infant to cardiac monitor for continuous monitoring.

4.12. Toes, feet and buttocks should be inspected for signs of circulatory compromise, if umbilical arterial or venous catheter is used for blood exchange.

4.13. Strict input and output of blood should be measured and recorded including timing.

4.14. Blood warmer should be set at 37 degrees centigrade and allow the blood to run through the blood warming device, but should not exceed 39 degrees centigrade.

4.15. Do not store blood in unit refrigerators. If unable to start blood exchange within 30 minutes after the blood is taken from the Blood bank, return the blood immediately for safe storage. In case of unavoidable circumstances that blood is not given 30 minutes after taking from the blood bank, OVR must be written, and blood should be returned to Blood bank.

4.16. Blood exchange should be stopped immediately if signs of blood reaction occur.

5.0 PROCEDURE (S)

5.1. Infant will be kept NPO (nothing per orem) for 3 - 4 hours before the procedure or the stomach contents will be aspirated with a nasogastric tube (NGT) if not NPO.

5.2. Request a type and cross match blood as ordered (RBC group O negative or infant's blood type).

5.3. Plug in the blood warmer set with the blood warming tubing inserted.

5.4. Set the blood warmer at 37 degrees centigrade and allow the blood to run through the blood warming device, but should not exceed 39 degrees centigrade.

5.5. Prepare the normal saline flush with 1 "u" of Heparin per 1 ml, as ordered by Physician.

5.6. Place infant under radiant warmer on servo control & keep temperature with thermos-neutral zone. Hypothermia increased oxygen & glucose consumption causing metabolic acidosis & also inhibit the binding capacity of albumin and bilirubin and hepatic enzymatic reaction thus, increase risk of kernicterus. Hyperthermia damages the donor erythrocytes elevating the free potassium content thus predisposing the infant to cardiac arrest.

5.7. Attach electronic cardiac monitoring device and pulse oximeter to infant. Gently restrain the extremities.

5.8. Obtain pre-transfusion vital signs, temperature and abdominal girth. Vital signs including temperature will be recorded every 15 minutes throughout the exchange. Notify the physician of any heart abnormalities, color changes or agitation.

5.9. Obtain pre-exchange lab work including serum total bilirubin, CBC with platelets, electrolytes, calcium, glucose, and coagulation profile and blood gas.

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5.10. Obtain glucose value at bedside using the glucometer.

5.11. Blood brought by registered nurse from blood bank must be counter checked with the doctor/nurse to confirm the order, amount and stability of cross matched blood before infusion:

5.11.1. Serial Number of blood

5.11.2. Expiry date

5.11.3. Blood Group

5.11.4. Negative Serology Reports

5.11.5. Check pack red blood cells for clots and hemolysis, and punctured blood bag.

5.12. Confirm the identity of the patient and the blood by a staff nurse and/or a physician before initiating the blood exchange. All information must match the medical record and the patient, to confirm correct blood is given to the right patient. Do not proceed with the procedure if there is discrepancy. Contact blood bank immediately.

5.13. **Assisting with the Exchange Transfusion**:

5.13.1. Agitate the blood/invert the bag every 15 minutes to avoid sedimentation of the RBCs.

5.13.2. Tape the blood collection bag securely to the bedside.

5.13.3. Withdraw blood specimen from unit of blood to check potassium level at the beginning of the transfusion as ordered.

5.13.4. Assist with placement of the umbilical lines if not previously placed. Exchange transfusions can be done through a single central line or through two central lines (UAC and UVC). The set up will vary depending on what type and number of lines that are used.

5.13.5. Open the exchange transfusion tray using aseptic technique.

5.13.6. Assist the doctor in setting up blood exchange transfusion maintaining strict sterile technique, connecting to appropriate line. A 4-way stopcock, if available, is used which allows a clockwise rotation in the order used:

5.13.6.1. Port 1 is used to withdraw blood from the patient and infuse blood into the patient. It is usually connected to the umbilical catheter.

5.13.6.2. Port 2 is used to discard old blood, and is connected to the tubing entering the waste collection container.

5.13.6.3. Port 3 is used to draw up the new blood and is connected to the blood warmer tubing.

5.13.6.4. Port 4 is used as an injection port, and is attached to a syringe to allow injection of medications to the patient, if needed.

5.13.6.5. If only 3-way stopcocks are available, connect two together using aseptic technique: one port connected to the blood warmer tubing, opposite port connected to waste collection bag, forward port attached to umbilical line and rear port connected to 10 or 20 ml syringe.

5.13.7. Record time and amount of each pass on the Newborn exchange transfusion record.

5.13.8. The physician will start the exchange transfusion under strict aseptic technique, will draw up blood from the donor bag after properly positioning the stopcock, and then transfuse to the newborn. The physician will announce the amount "in" to the nurse to be recorded.

5.13.9. Amount withdrawn is announced by the physician as blood "out" and is recorded as such, and is ejected into the discard bag.

5.13.10. The procedure will be repeated until the appropriate volume is exchanged. The physician will announce each amount "in" and "out". Average time between passes "in" and "out" is 3 – 5 minutes or approximately 2-3 ml/kg/min of blood removed / replaced. The exchange/replacement is done in 10 – 20 ml aliquots.

5.13.11. As the procedure goes on, the nurse may notice "sticking" of the stopcock valves that will slow blood flow from the patient, gently flushed with heparinized saline to clear the "sticking" and improves the smoothness of blood flow withdrawn from the patient.

5.13.12. During the exchange transfusion, the blood should be regularly agitated to keep the red blood cells from settling.

5.13.13. Observe for tremors. Send blood works for hypocalcaemia and glucose after each 40 ml/kg exchanged.

5.13.14. Observe for chills, flashing, dyspnea, rash or other signs of transfusion reaction, (Refer to policy of Blood Transfusion Reaction (MPP-LB-002)

5.13.15. Stop blood transfusion immediately for any adverse hemolytic reaction:

5.13.15.1. Assess the neonate

5.13.15.2. Assist the doctor to stabilize the neonate

5.13.15.3. Prioritize your nursing intervention according to your assessment.

5.13.15.4. Inform the CN/HN/Supervisor

5.13.15.5. Complete the Patient information portion on the Transfusion Reaction Form & return to the Blood bank along with blood component bag, transfusion set and collected samples

5.13.16. Collect blood sample in plain tube and in Edta tube, and first urine sample after blood reaction, to be forwarded to the Blood bank.

5.13.17. Write an OVR, to be forwarded thru proper channel immediately then to Patients' Safety.

5.14. **After the Exchange Transfusion**:

5.14.1. When the exchange is completed, clear the lines with normal saline and change the stopcock (3 or 4 way).

5.14.2. Replace maintenance fluids to the line or keep vein open as ordered.

5.14.3. Remove drapes and restraints.

5.14.4. Discard used syringes according to waste disposal, and blood bag to be returned to Blood bank.

5.14.5. Continue monitoring and recording of vital signs every 15 minutes' x 1 hour, every 30 minutes' x 2 hours and every hour x 2 hours, then follow NICU monitoring.

5.14.6. Infant should remain NPO for at least 4 hours. Monitor abdominal girth and bowel sounds every 3 – 4 hours when the exchange has been done through umbilical lines. Observe for signs of feeding intolerance, including necrotizing enter colitis.

5.14.7. The toes, feet and buttocks of the infant should be inspected for signs of circulatory compromise, report to physician immediately.

5.14.8. One-hour post exchange, check the blood glucose, then monitor levels every 2 – 4hours or as ordered for 24 hours.

5.14.9. Two hours post exchange send blood for total serum bilirubin, CBC with platelets and ionized calcium as ordered, then as clinically indicated.

5.14.10. Repeat blood gas as ordered, then as clinically indicated.

5.14.11. Accurate documentation to provide a comprehensive view of the procedure and to evaluate the effectiveness to the condition of the newborn:

5.14.11.1. Date & time blood exchange started and completed.

5.14.11.2. Volume of blood withdrawn and infused and total volume exchanged.

5.14.11.3. Initial & subsequent vital signs, abdominal girth and results of lab works obtained before, during and after the procedure.

5.14.11.4. Name of the physician who performed the procedure.

5.14.11.5. The new-born's clinical status before, during and after the procedure and the new-born's tolerance of the procedure.

6.0 EQUIPMENTS

6.1. Resuscitation & Suction equipment, Crash cart

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6.2. Sterile gown & gloves, masks, caps, sterile drapes or towels
6.3. Cardiopulmonary monitor
6.4. Exchange Blood transfusion set
6.5. Blood from blood for exchange transfusion as ordered
6.6. Double volume exchange = 160 ml/kg
6.7. Single volume exchange = 80 ml/kg
6.8. Rh Incompatibility: use O negative compatible blood
6.9. Sepsis: Infant's blood type or O negative
6.10. O negative whole blood may be used in any emergency
6.11. Three-way stopcock including four-way stopcock
6.12. Glucometer for blood glucose
6.13. Blood warmer
6.14. 10 or 20 ml syringes and 18 gauge needles
6.15. Pre- heated radiant warmer with servo control
6.16. Chlorhexidine swabs
6.17. Heparin dilution
6.18. Umbilical catheter tray if lines have not been previously placed
6.19. Saline solution for partial exchange
6.20. Waste blood receptacle

7.0 ATTACHMENT/FORMS

7.1. Blood Exchange Transfusion Record
7.2. Consent for Blood / Blood Components Transfusion (GDOH-COR-CBCT-357)
7.3. NICU Flow Sheet (GDOH-NUR-NIF-199)
7.4. Nurse Observation Form (Form No. 43)
7.5. Nurses Progress Notes (GDOH-NUR-NPN-203)

8.0 REFERENCES

8.1. Guidelines for Neonatal Care (NICU) MOH Saudi Arabia

8.2. Hematology of Infancy and Childhood, 6th edition by Nathan, Orkin, Girsburg and Look

8.3. Perfecting Clinical Procedures, 2008 by Lippincott Williams & Wilkins

8.4. http://www.gosh.nhs.uk/health-professionals/clinical-guidelines/exchange-transfusion-protocol

8.5. http://en.wikipedia.org/wiki/Exchange_transfusion

9.0 REVISION CLAUSE

9.1. Modification of policy format, lay out and policy coding

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____/____/14____ H ____/____/20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

CONSENT FOR BLOOD/BLOOD COMPONENTS TRANSFUSION موافقة على نقل الدم/ مشتقات الدم

I, the undersigned, _____ in my Capacity as

☐ **Patient** ☐ **Legal Guardian** ☐ **Relative**

(Specify: _____) of

(Patient's Name: _____)

That I have ..... I need the

Following units of .....

Since this will support the treatment of my condition.

It has been explained to me about the:

- Nature of the procedure.
- Benefits of this procedure.
- Risks involved with this procedure.
- Potential risks of not carrying out this procedure.
- Possible alternative treatment modalities.

I knowledge that all relevant information about the procedure has been given to me in the language that I understand,

I have the opportunity to ask questions concerning me / my patient's condition and about the procedure and all questions have been answered to my satisfaction.

Patient's/Guardian's Name: .....

Relationship to patient: .....

Signature;

Date:...../...../.....

I have seen this consent before starting blood/ blood components transfusion and explained this to patient/ guardian.

Physician's Name: .....

Signature:

Stamp:

Witness's Name: .....

Signature:

Date:...../...../.....

أنا الموقع أدناه: _____ بصفتي

☐ **المريض/** ☐ **ولي الأمر/** ☐ **القريب**

صلة القرابة:(_____)

اسم المريض:(_____)

اعاني من .....

وحاجتي إلى الوحدات التالية من .....

وذلك حسب حالتي المرضية كون هذا الدم سيساعد في التغلب على مرضي.

لقد تم الشرح التام لي عن مايلي:

- طبيعة الاجراء .
- الفوائد المرجوة من هذا الاجراء .
- المخاطر المرتبطة بهذا الاجراء .
- المخاطر المترتبة على عدم عمل هذا الاجراء .
- البدائل الممكنة للعلاج .

وأقر بأن كافة المعلومات الهامة ذات الصلة بهذا الإجراء قد تم ايضاحها لي بلغة مفهومة من قبلي، كما اتيحت لي الفرصة لطرح الاسئلة التي تخطر لي / لمريضي، وأنه قد تمت الإجابة على جميع استفساراتي بصورة مرضية.

إسم المريض/ولي الأمر :.....

صلة القرابة: .....

التوقيع:

التاريخ:...../...../.....

اطلعت على هذا الإقرار قبل إجراء نقل الدم ومشتقاته وقمت بشرح ذلك للمريض/ولي أمره.

اسم الطبيب: .....

التوقيع :

الختم :

اسم الشاهد: .....

التوقيع:

التاريخ:...../...../.....

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وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: _____ رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

BLOOD & BLOOD PRODUCTS REQUEST & RELEASE FORM

Note: One form must be submitted for each unit ordered. Blood release copy remains on chart until blood is needed. Blood bank copy must be returned to blood bank on completion of transfusion or return of blood.
RETURN BLOOD TO CENTRAL BLOOD BANK WITH 30 MINUTES FROM RELEASE OF BLOOD FROM BLOOD BANK IF NOT TRANFUSED.

Requisition for: ☐ Packed Red Cells ☐ Whole Blood ☐ Fresh Frozen Plasma ☐ Cryoprecipitate ☐ Platelet Concentrate

Special Instructions: ☐ Irradiated ☐ Leukocyte poor (Filtered) ☐ Volume _____ml

Requesting Physician's name: _____ Stamp&Signature: _____ Date:___/___/ ___

RELEASE BLOOD PRODUCTS

Unit No _____ Product _____ Blood Group _____ Expiry Date:___/___/ ___

Blood unit issued by: Name _____ Sign _____ Date:___/___/ ___ Time _____

Blood unit received by: from blood bank: Name: _____ Signature: _____

Blood unit received in the ward by: Name: _____ Signature: _____ Time: _____

TRANSFUSION RECORD

(This part must be completed when transfusion is started)

Pre transfusion

Patient's temp. _____ BP _____ Pulse _____

Date of transfusion _____ Time _____

I have verified ID, Blood Group and Rh of the recipient and donor on bag label and blood issue and compatibility testing forms. I have further verified that the recipient is same person named on all documents.

Remarks: _____

_____
Name and signature of person starting transfusion

_____
Name and signature of 2nd verifying person

(This part must be completed when transfusion is stopped)

Post transfusion

Ending time: _____ Amount given _____

Patient's temp. _____ BP _____ Pulse _____

_____
Name and signature of person completing transfusion

Transfusion Reaction: ☐ None

☐ Yes (Specify)_____

If transfusion reaction is suspected then immediately:

1. Discontinue transfusion.
2. Treat shock, if present.
3. Notify attending physician.
4. Notify blood bank at once and send to blood bank/ lab.
   a. Unused blood, transfusion set with tubing attached (remove needle)
   b. Recipient venous blood sample 5ml EDTA tube.
   c. Completed transfusion reaction form.
   d. Next voided urine to central lab.

Other difficulties: (Equip, clots etc) ☐ None

☐ Yes (Specify)_____

(This part must be completed if blood is returned to BB)

Reason for return _____

Date ___/___/ ___ Time _____

Returned by: Name and signature _____

Received in blood bank :Date ___/___/ ___ Time ___

By Name and Sign _____

(I verify temperature and integrity of bag is acceptable)
If unacceptable for reuse, fill incident report form.

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MRN: |_|_|_|_|_|_|_|_|_|_| رقم الملف الطبي:

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Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NEONATAL ICU FLOW SHEET

DATE	ADMISSION DATE	DATE OF BIRTH	AGE (IN DAYS)	DAYS IN NICU	GESTATIONAL AGE	CORRECTED AGE
BIRTH WEIGHT	**PRESENT WEIGHT**	**PREVIOUS WEIGHT**	**CHANGE**	**LENGTH**	**HEAD CIRCUMFERENCE**	**BLOOD GROUP**

DIAGNOSIS:

SAFETY CHECKLIST	0700		1500		2300		STATUS	RN INITIAL
ALARM LIMITS	Low	High	Low	High	Low	High		0700H
Cardiac								
Respiratory								
SaO2								1500H
TcPO2								
TcPCO2								
Mean Arterial Pressure								2300
Central Venous Pressure								

TUBE PLACEMENT			
TYPE	DATE	SIZE	LEVEL
ETT			
UAC			
UVC			
OGT/ NGT			
PICC/ CL			

EQUIPMENT	Morning	Evening	Night
O2 /bagging unit			
Suction apparatus			
Chest tube H2O pressure			
ID band			
Doctor's order			
Medication Sheet			
IV/IA fluids, drug infusions, rates, sites including heplocks			
Phototherapy lights/Eye pads			
Ventilator changed: Date: Due:			
Isolette changed: Date: Due:			
ETT Suction Done: Due:			
Dressing Done: Due:			

SURGICAL PROCEDURES	
1.	DATE
2.	DATE
3.	DATE
4.	DATE

PARENT'S VISIT	
MOTHER:	EBM
FATHER:	LAST VISIT:
OTHERS:	

24 HOUR INTAKE

	TYPE	AMOUNT
ORAL/ NGT		
TPN	Amino acids	
	Lipids	

24 HOUR OUTPUT

Urine: Total ml per 24hours		ml/kg/hour.
Other losses:		
Stool/ Ostomy: Total ml per 24hours		Last stool:
Gastric drainage: Total ml/24hours		
Chest tube drainage: Total ml per 24hours		
Blood loss: Total ml per 24hours		Total output since transfusion:

TOTAL INTAKE		TOTAL OUTPUT		BALANCE	

24 HOUR SUMMARY:

ORDERED INTAKE: (ml/kg/day)	ACTUAL INTAKE: (ml/kg/day)	CALORIES: kcal/kg/day

0 000000 001991

SN

Name:_____ الاسم:

MRN: |_|_|_|_|_|_|_|_|_|_| رقم الملف الطبي:

TIME	CNS ASSESSMENT						TEMPERATURE				CARDIOVASCULAR ASSESSMENT						
	ENVIRONMENT	STATE	ACTIVITY	FONTANELLE	TONE	POSITION	SET	AIR	HUMIDITY %	BABY	HEART RATE	SYS/ DIAS	MEAN (BP)	BP SITE	CVP	PULSES	COLOR
0800H																	
0900H																	
1000H																	
1100H																	
1200H																	
1300H																	
1400H																	
1500H																	
1600H																	
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2200H																	
2300H																	
2400H																	
0100H																	
0200H																	
0300H																	
0400H																	
0500H																	
0600H																	
0700H																	

ENVIRONMENT

C- cot
I-isolette

W-Warmer
CB- Crib

STATE

C-comatose
QS-quiet sleep
AS-active sleep
QA-quiet alert
AA-active alert
Cr-crying
S-sedated

ACTIVITY

P- paralyzed
U-unresponsive
L- lethargic
A- appropriate
J – jittery
S- seizures
I- irritable

FONTANELLE

S-soft
F-flat
D-depressed
O-overriding sutures
F-full
SP-splayed sutures

TONE

Fl-flaccid
❒ H-hypertonic
❒ H-hypotonic
√-appropriate

PULSES

0-absent
1-difficult to palpate
2-soft/easy to palpate
3-slightly full
4-bounding/ full
=-equal

COLOR

P-pink
W-pale
D-dusky
R-ruddy
A-acrocyanosis
C-cyanotic
J-jaundiced
M-mottled

ACCEPTED PARAMETERS

Heart Rate-
O_2 sat –
Mean BP -

0 000000 001991

SN

Name:_____ الاسم: | MRN: رقم الملف الطبي:

RESPIRATORY MONITORING

TIME	RR	BS/ A/E	Effort	O2 Sat.	Probe change (Site)	CT #1	CT # 2	O2 L/min	*Pain Score	VENTILATOR DATA											SUCTION		
										Mode	Rate	AMP	MAP	PIP/PEEP	VG	PS/ PG	I.E	FiO2	FLOW	iNO	Instill	ETT	Nares/ PO
0800H																							
0900H																							
1000H																							
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Breath Sounds
Cl- clear
Cr- crackles
W- wheezes
Rh- rhonchi
S- stridor

Air Entry
= equal
❒-decreased
Define lobe-(LLL/ LUL/RLL/RML/RUL)

Effort
e - easy
ss- substernal
sc- subcostal
ic- intercostal
nf- nasal flaring
gr- grunting
tt- tracheal tug

O2 Therapy
Hood
Nasal cannula
Mask
Incubator

*** Refer to assessment guide**
*Indicate the type of tool used for pain.
*If pain present, make use of pain assessment and management sheet.

Mode
CV-conventional ventilator
HFOV-high frequency oscillation vent
CPAP-continuous positive airway pressure
NP-nasal prong CPAP

CT'S
B-bubbling
F-fluctuating
C-clamped
❒-no activity

Suction
Consistency
ls- loose
th-thick
p- plugs

Color
cr- creamy
cl- clear
wh- white
y- yellow
g- green
s- sanginous

Instill
0.25ml
0.5ml
1.0ml

Amount
sc- scant
sm- small
m- moderate
l- large
c- copious

0 000000 001991 SN

Name:_____ :الاسـم

MRN: | | | | | | | | | | :رقم الملف الطبي

TIME	BLOOD GASES								ABDOMEN		ASSESSMENT		Blood Glucose	SBR	Phototherapy
	Site/ Source	PH	PCO2	PO2	HCO3	BE	Sat	Metal Hb	Bowel Sounds	Abdominal Girth	*Fall Risk Score	*Skin Risk Score			
0800H															
0900H															
1000H															
1100H															
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Bowel Sounds

❒-present
❒- increased
❒-decreased
❒-absent

Abdominal girth

s-soft
f-flat
fl-full
d-distended
fm-firm
bl-bowel loops

***Refer to assessment guide**
Indicate the type of tool used.

0 000000 001991 SN

Name:_____ الاسم:	MRN: رقم الملف الطبي:

INTAKE

Intravenous Infusion/Medications/Blood Products

Time	#1		#2		#3		#4		#5		#6		#7		#8		#9		Fluh	Meds	√
	IV	TOTAL	IV	TOTAL	IV	TOTAL	IV	TOTAL	IV	TOTAL	IV	TOTAL	IV	TOTAL	IV	TOTAL	IV	TOTAL			
0800H																					
0900H																					
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0800H																					
0900H																					
Total / 24 hors																					

Feeds

Type	Amount	Route	Increment

Name:_____ الاسم:	MRN: رقم الملف الطبي:

OUTPUT

TIME	GASTRIC ASP	VOMITUS	DIALYSIS	BLOOD LOSS	BOWEL	URINE	PH	SG	GLUCOSE	PROTEIN	LEUKO	BILIRUBIN	DRAIN					
													1	2	3	H/O	TOTAL	BAL.
0800H																		
0900H																		
1000H																		
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0500H																		
0600H																		
0700H																		
TOTAL 24 HRS																		

SPECIAL INSTRUCTIONS	FOLLOW UP LABORATORY RESULTS & PROCEDURES

Legend: H/O – Hourly Output; Bal - Balance

0 000000 001991 SN

Name:_____ الاسم:	MRN: | | | | | | | | | | رقم الملف الطبي:

NURSES PROGRESS NOTES

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

Name:_____ الاسم:	MRN: رقم الملف الطبي:

NURSES PROGRESS NOTES

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

Ministry of Health
General Directorate of Health Affairs
Al-Baha

وزارة الصحة
المديرية العامة للشئون الصحية بمنطقة الباحة

HOSPITAL :

Health Care No .: ☐☐☐☐☐☐☐☐☐☐☐☐

Name : .....

Age : ☐☐ Yrs./Month Sex : ☐M ☐F

Nationality : .....

Consultant In-Charge : .....

Dept : ..... Unit : .....

NURSES OBSERVATION FORM نمـوذج مـلاحظـات التمـريضـ

Nurse / Date	Time	Temp.	Puls Rate	Resp Rate	B.P.	Pupils		Level of Consciousness Muscle Tone &Reflex	Cough Reflex	Response to Stimuli		Perception Pulses			Skin Color	Rema-rks	Sign.
						R	L			Verbal	Painful	ORAL	Audio Aural	visual			

SPECIFIC NURSING CARE COMMENTS :

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KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

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SN